ECON 0100 | Demo C

Solutions

Answers and work are shown in the red blocks.

Patronus Lessons

Every winter, dementors drift through Hogsmeade, and the surest defense against them is the Patronus Charm. Private tutors sell lessons in the charm. Each witch or wizard who learns it also protects their neighbors from dementors, an external benefit of 10 coins per lesson. Prices are in coins and quantity is in lessons. The Demand (Marginal Private Benefit) and Supply (Marginal Cost) equations for lessons are the following:

$$D : P_b = 40 - Q_d \qquad S : P_s = 10 + Q_s$$

C1.2 | Patronus Lessons

The graph shows the market for lessons: marginal private benefit (MPB), marginal social benefit (MSB), and supply (S). The lettered regions are for the questions.

MSB lies _____ coins above MPB at every quantity
Market Quantity ($\hat{Q}$): _____
Efficient Quantity (Q^*): _____
Deadweight Loss (*letter*): _____
Deadweight Loss (*area*): _____
MSB of the Last Lesson Traded: _____
MC of the Last Lesson Traded: _____
The Market Trades: ☐ too many ☐ too few lessons

MSB lies 10 coins above MPB. Market Quantity $\hat{Q}$: 15 lessons. Efficient Quantity Q^*: 20 lessons. Deadweight Loss Region: B. Deadweight Loss: 25 coins. Last lesson traded: $MSB = 35$, $MC = 25$. The market trades too few lessons.

Each lesson is worth its buyer MPB and another 10 to the neighbors it protects, so MSB is MPB shifted up by 10 at every quantity:

$$MSB = MPB + 10 = 50 - Q$$

Buyers count only their own benefit, so the market trades where MPB meets supply:

$$40 - Q = 10 + Q \quad \Rightarrow \quad \hat{Q} = 15, \quad \hat{P} = 25$$

Efficiency counts the neighbors too, so it needs MSB to meet supply:

$$50 - Q = 10 + Q \quad \Rightarrow \quad Q^* = 20, \quad P = 30$$

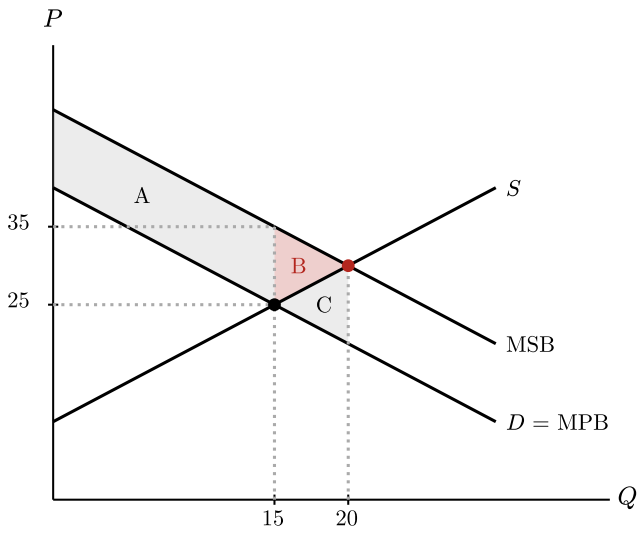


Figure 1: Patronus lessons with region B, the deadweight loss, in red

The deadweight loss is B, the triangle between MSB and supply over lessons 15 to 20. Each of those lessons is worth more to society than it costs, yet none is traded: at $Q = 15$ the last lesson has $MSB = 35$ and $MC = 25$ (read off the price axis), so one more lesson would add 10 more in benefit than it costs, and the market trades too few. The height at $Q = 15$ is the external benefit of 10 and the base is the 5 lessons the market skips:

$$\text{DWL} = \tfrac{1}{2} \times 10 \times (20 - 15) = 25$$

A is the external benefit the 15 traded lessons already deliver. C, between supply and MPB, is how far each of lessons 16 to 20 costs more than its buyer alone would pay, which is why the market skips them; it is not a loss to society.

C2.2 | A Subsidy for Lessons

To encourage lessons, the Ministry pays tutors a subsidy of 8 coins per lesson. For this question, ignore the external benefit: demand is the marginal benefit (MB) of a lesson.

Buyer's Price (P_b): _____
Seller's Price (P_s): _____
Subsidy Quantity (Q^{Sub}): _____
Deadweight Loss (DWL): _____
Last Lesson Traded, MC vs. MB: ☐ above ☐ below

Buyer's Price P_b: 21 coins. Seller's Price P_s: 29 coins. Subsidy Quantity Q^{Sub}: 19 lessons. Deadweight Loss: 16 coins. For the last lesson traded, MC is above MB.

Ignoring the external benefit, demand is the marginal benefit, and the market without the subsidy is efficient: $40 - Q = 10 + Q$ gives $Q = 15$ at $P = 25$.

A subsidy is the tax wedge turned around: the seller now receives 8 more than the buyer pays, so the wedge sits to the *right* of 15:

$$P_s = P_b + 8 \quad \Rightarrow \quad 10 + Q = (40 - Q) + 8 \quad \Rightarrow \quad Q = 19$$

so $P_b = 40 - 19 = 21$ and $P_s = 10 + 19 = 29$.

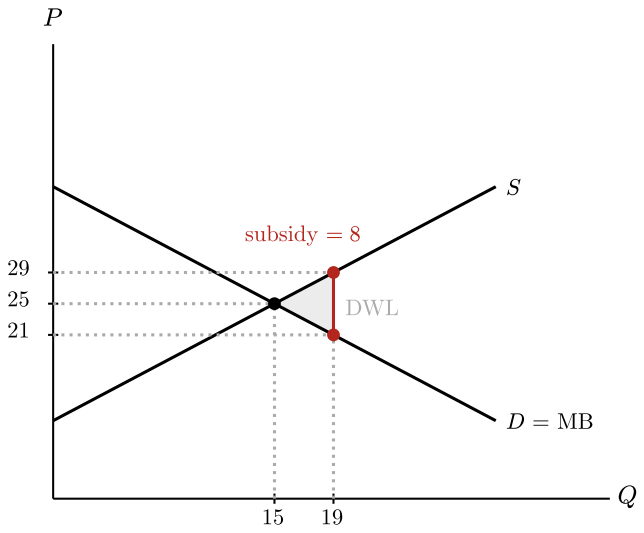


Figure 2: An 8-coin subsidy on Patronus lessons, ignoring the external benefit

At $Q = 19$, the last lesson costs $MC = 29$ to provide but is worth only $MB = 21$ to its buyer, so MC is above MB. The same is true of every lesson from 16 to 19: each trades only because the Ministry pays the difference. The triangle between supply and demand over those 4 lessons is the deadweight loss:

$$\text{DWL} = \tfrac{1}{2} \times 8 \times (19 - 15) = 16$$

Counting the external benefit of 10 is a corrective-policy question (C3.1): the efficient quantity becomes 20, and a subsidy of exactly 10 would reach it.

Flummoxed by the Floo Powder Fiasco

Floo Powder is a magical commodity used by witches and wizards to transport themselves through fireplaces to different locations. While it's often convenient, its use is associated with negative environmental effects, including air pollution and the depletion of magical resources used to produce the powder. The Ministry's Bureau of Economic Analysis has estimated the externality to be 12 coins per tin with the following supply and demand equations:

$$D : P_b = 120 - 2Q_d \qquad S : P_s = Q_s$$

C1.1 | The Fiasco

Start by using a graph and algebra to describe the market failure.

Graph demand, supply, and MSC, and shade the deadweight loss.
Marginal Social Cost (MSC): _____
Market Quantity ($\hat{Q}$): _____
Efficient Quantity (Q^*): _____
Deadweight Loss (DWL): _____
The Market Trades: ☐ too much ☐ too little

Marginal Social Cost: $MSC = 12 + Q$. Market Quantity $\hat{Q}$: 40 tins. Efficient Quantity Q^*: 36 tins. Deadweight Loss: 24 coins. The market trades too much.

The sellers' marginal cost is the supply curve, $P = Q$. Each tin also costs everyone else 12, so marginal social cost is supply shifted up by 12:

$$MSC = 12 + Q$$

The market ignores the external cost and trades where demand meets supply:

$$120 - 2Q = Q \quad \Rightarrow \quad \hat{Q} = 40, \quad \hat{P} = 40$$

Efficiency needs demand (marginal benefit) to meet MSC:

$$120 - 2Q = 12 + Q \quad \Rightarrow \quad Q^* = 36, \quad P = 48$$

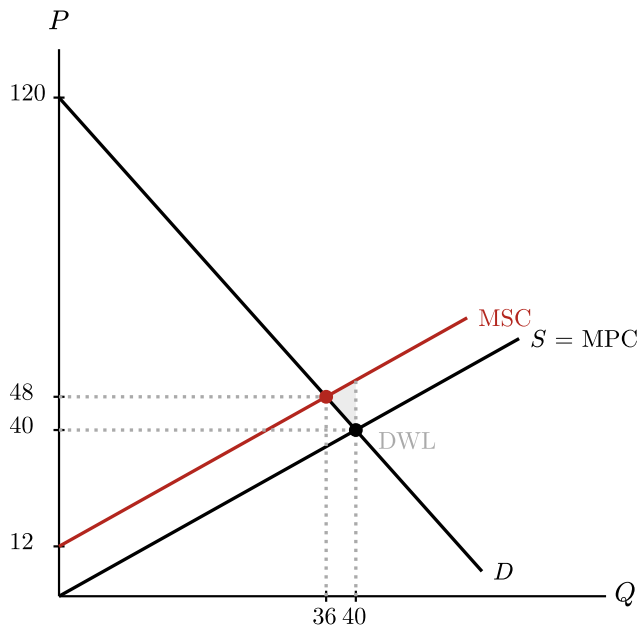


Figure 3: Floo Powder market failure with the deadweight loss shaded

The deadweight loss is the triangle between MSC and demand over tins 36 to 40, with corners $(36, 48)$, $(40, 52)$ and $(40, 40)$. Its height at $Q = 40$ is the external cost of 12 and its base is the 4 extra tins, so DWL $= \frac{1}{2} \times 12 \times (40 - 36) = 24$.

At $Q = 40$, $MSC = 12 + 40 = 52$ and $MB = 120 - 80 = 40$. The last tin costs society 12 more than it is worth to its buyer, and the same is true of every tin from 36 to 40, so the market trades too much.

C2.1 | A Tax for Revenue

Needing revenue, and not yet knowing the size of the externality, the Ministry imposes a tax of 6 coins per tin on Floo Powder. For this question, ignore the external cost.

Buyer's Price (P_b): _____
Seller's Price (P_s): _____
Post-Tax Quantity (Q^{Tax}): _____
Tax Incidence on Buyers: _____
Tax Incidence on Sellers: _____
Deadweight Loss (DWL): _____
Who Bears More of the Tax? ☐ buyers ☐ sellers ☐ equal

Buyer's Price P_b: 44 coins. Seller's Price P_s: 38 coins. Post-Tax Quantity Q^{Tax}: 38 tins. Tax Incidence: buyers 4, sellers 2. Deadweight Loss: 6 coins. Who bears more: buyers.

Before the tax, buyers and sellers face one price, $P = 40$ at $Q = 40$ (C1.1). With the tax, the buyer price sits 6 above the seller price:

$$P_b = P_s + 6 \quad \Rightarrow \quad 120 - 2Q = Q + 6 \quad \Rightarrow \quad Q = 38$$

so $P_b = 120 - 76 = 44$ and $P_s = 38$. Check: $44 - 38 = 6$.

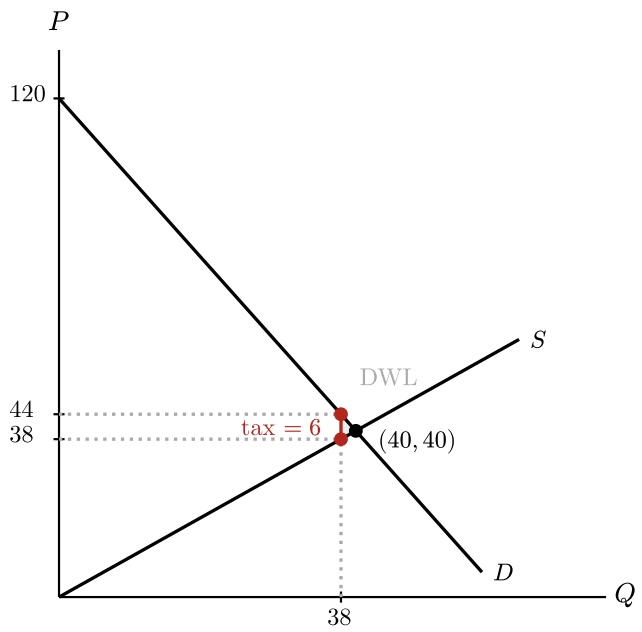


Figure 4: A 6-coin tax on Floo Powder

Incidence: buyers pay $44 - 40 = 4$ more than before and sellers receive $40 - 38 = 2$ less, and the two add to the 6 tax. Buyers bear more because demand is steeper than supply (slope 2 against 1): buyers are the side less able to move away from the tax.

Ignoring the external cost, the untaxed market's 40 tins were efficient, so the tax destroys the surplus on the 2 tins it prevents, the triangle between demand and supply from 38 to 40:

$$\text{DWL} = \tfrac{1}{2} \times 6 \times (40 - 38) = 6$$

Counting the external cost of 12 turns this around: those 2 tins cost society more than they were worth, so the tax moves the market *toward* the efficient 36. That is the C3.1 question.

C3.1 | Designing the Policy

The Ministry of Magic has hired you to design a policy that corrects the market failure caused by the negative externality of Floo Powder use. Propose the policy you recommend and how it would impact the market for Floo Powder and its users.

Policy Type: _____
Policy Size: _____
Post-Policy Quantity (Q^{τ}): _____
Post-Policy Deadweight Loss: _____
Post-Policy Quantity vs. Efficient: ☐ above ☐ at ☐ below

Policy Type: a per-unit tax. Policy Size: 12 coins per tin. Post-Policy Quantity Q^{τ}: 36 tins, at the efficient quantity. Post-Policy Deadweight Loss: 0.

The externality is a cost, so the policy is a tax, and its size is the external cost per tin, 12. Sellers then pay the full social cost: supply plus the tax is $12 + Q$, exactly MSC. The buyer price sits 12 above the seller price:

$$P_b = P_s + 12 \quad \Rightarrow \quad 120 - 2Q = Q + 12 \quad \Rightarrow \quad Q = 36$$

so $P_b = 120 - 72 = 48$ and $P_s = 36$.

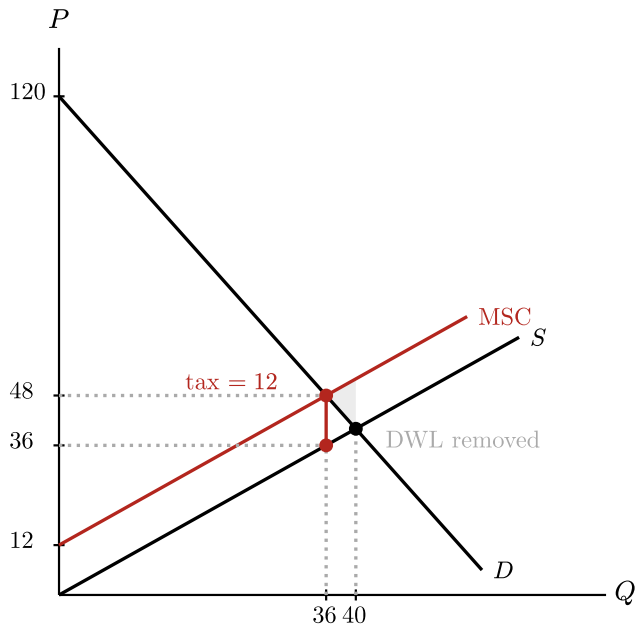


Figure 5: A 12-coin corrective tax on Floo Powder

Without a policy the market trades 40 tins. The tax moves it down to 36, exactly the efficient quantity, so the deadweight loss from C1.1 is gone. Users of Floo Powder pay 48 instead of 40 and buy 4 fewer tins; sellers keep 36 instead of 40. (The Ministry collects $12 \times 36 = 432$ coins; not required.)

The 6-coin tax in C2.1 moved quantity from 40 to 38, the right direction but not far enough: 38 is still above the efficient 36, because the tax is smaller than the external cost, and it leaves a deadweight loss of $\frac{1}{2} \times 6 \times (38 - 36) = 6$, the gap between MSC (50) and demand (44) at 38. The tins from 36 to 38 still cost society more than they are worth.